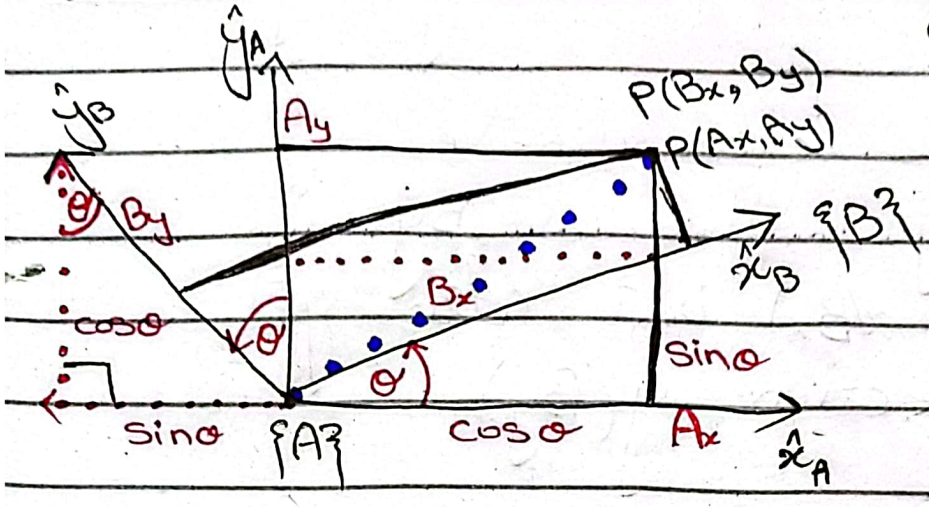
$\sin\theta = \frac{y}{r}$ / opposite / hypotenus

$\cos\theta = \frac{x}{r}$ / adjacent / hypotenus

$\tan\theta = \frac{y}{x}$

$\hookrightarrow \frac{\sin\theta}{\cos\theta} = \frac{y}{r} \cdot \frac{r}{x}$

# 2D Rotation Matrix about origin at angle $\theta$




∵ Hypotenus is always opposite to right angle.

$$A_P = A_x \hat{x}_A + A_y \hat{y}_A \quad \text{——} (1)$$

$$B_P = B_x \hat{x}_B + B_y \hat{y}_B \quad \text{——} (2)$$

$$\hat{x}_B = \sin\theta \hat{y}_A + \cos\theta \hat{x}_A$$

$$\hat{y}_B = -\sin\theta \hat{x}_A + \cos\theta \hat{y}_A$$

∵ The horizontal component of $\{B\}$ ie $x_B$ after rotation by $\theta$ horizontal component $B_x$ which is $\cos\theta$ and $B_y$ which is $-\sin\theta$

Effect on x and y Projection

The new x-coordinate is primarily influenced by the original $x(B_x)$ but also gets a projection $y(B_y)$.

Since the vertical displacement contributes in the negative x-direction, this introduce -ve sine. $(-B_y \sin\theta)$

$B_x$ contributes +vely to the new-x coordinate through $\cos\theta$

$B_y$ contributes -vely to the new x-coordinate due to downward projection when rotating.

* The sin and cos terms represents the original coordinates onto the new axis defined by the rotation angle $\theta$

* The terms $-B_y \sin\theta$ and $B_x \cos\theta$ reflect the transformation of the original coordinates as the point is rotated, contributing to the opposite axis due to the 90° phase difference in sin and cos

Put $\hat{x}_B$ and $\hat{y}_B$ in (2)

$$B_p = B_x(\sin\theta \hat{y}_A + \cos\theta \hat{x}_A) + B_y(-\sin\theta \hat{x}_A + \cos\theta \hat{y}_A)$$

$$B_p = B_x \sin\theta \hat{y}_A + B_x \cos\theta \hat{x}_A - B_y \sin\theta \hat{x}_A + B_y \cos\theta \hat{y}_A$$

$$B_p = (B_x \sin\theta + B_y \cos\theta)\hat{y}_A + (B_x \cos\theta - B_y \sin\theta)\hat{x}_A \quad \text{(3)}$$

$$A_p = B_p$$

So, Compare (2) & (03)
(Because they share the origin, i.e Starting and ending point is same)

$$A_x = B_x \cos\theta - B_y \sin\theta$$

$$A_y = B_x \sin\theta + B_y \cos\theta$$

$$\begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix} \begin{bmatrix} B_x \\ B_y \end{bmatrix}$$

Rotation Matrix about origin at angle $\theta$

$$A_p = R^B_p$$

$$A_p = {}^A R_B \cdot {}^B p$$

To determine the base and
1 without explicit labels of
sine ($\sin\theta$) and cosine ($\cos\theta$)

1) Identify the angle of interest ($\theta$), in the triangle on the next page, $\theta$ is indicated in the bottom-left part of $\Delta$.

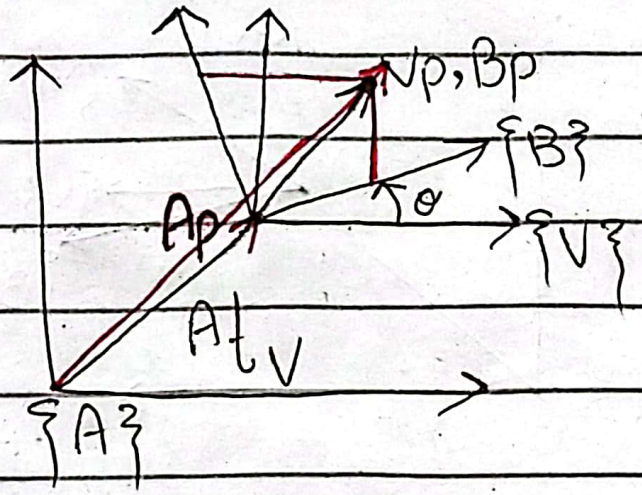
2) The side opposite to the $\theta$ is perpendicular. If $\theta$ is in bottom-left, the vertical side moving upward is typically 1. If $\theta$ is at bottom right, same as above explained.

3) Then Base & hypotenus will automatically detected.

Remember $\theta$ is not right angle.

# Transformation Matrix
## (Translation + Rotation)

### Translation in 2D




$$V_p = {}^{V}R_B \cdot B_p \quad \text{—— ①}$$

$$A_p = {}^{A}t_V + V_p \quad // \{v\} \cdot \| \{A\}$$

Put ① to $A_p$

$$A_p = {}^{A}t_V + {}^{V}R_B \cdot B_p$$

$$A_p = {}^{A}t_B + {}^{A}R_B \cdot B_p \quad // \{v\} \| \{A\}$$

$$\begin{Bmatrix} A_x \\ A_y \end{Bmatrix} = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix} \begin{Bmatrix} B_x \\ B_y \end{Bmatrix} + \begin{Bmatrix} A_x B \\ A_y B \end{Bmatrix}$$

$$A_x B = B\cos\theta$$
$$A_y B = B\sin\theta$$

$$\begin{bmatrix} A_x \\ A_y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos & -\sin & A x_B \\ \sin & \cos & A y_B \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ 1 \end{bmatrix}$$

Sign for Homogenous vector

$$\tilde{A_P} = \begin{bmatrix} {}^{A}R_B & {}^{A}t_B \\ 0,0 & 1 \end{bmatrix} \tilde{B_P}$$

$$\tilde{AP} = A_{t_B} \tilde{BP}$$

Explanation :-

* Homogenous Transformation Matrix

1) Ap refers to point with respect to A.
2) $A_{tV}$ refers ~~transformation~~ of Vector $\vec{V}$ with respect to $\vec{A}$.
3) Vp refers to point Rotation Matrix B with respect to A and point with respect to B.

~~$A_x B = A_x \cos\theta - A_y \sin\theta$~~

~~$A_y B = A_x \sin\theta + A_y \cos\theta$~~

- $P = \begin{pmatrix} a \\ b \end{pmatrix}$ $\tilde{P} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$

* Why 0 0 1 in Last Row?

The third row [0 0 1] in the matrix makes it a homogenous transformation matrix. Extra row and column are added to allow for translations (and other transformations like scaling). The extra 1 in the last row ensures that the matrix can include both rotation and translation in the same matrix operation (multiplication).

* $A_{tv} = A_xB, A_yB$

This refers to the new coordinates of vector A after the transformation (rotation and translation) relative to B.

Basically, $A_{+V}$ is the translation of $\vec{V}$ respect to $\vec{A}$

But, $A_{+V}$ would represent the displacement/movement of $\vec{V}$ respect to point A.

After applying translation, $\vec{V}$ is moved and new position is $A_{+V}$.

$A_{+V}$ is part of transformation process where you first translate $\vec{V}$ then perform additional operations like rotation or adjusting relative to another vector like Vp.

In summary, $A_{+V}$ is the translated version of vector $\vec{V}$ relative to point A and determining the final position of point Ap in transformation matrix.

# ORTHOGNAL / ORTHONORMAL

$A^T = A^{-1}$

↳ $AA^T = I$

$$AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Homogenous Matrix :